# Automatic Meeting Minutes & Action Item Extraction

An end-to-end NLP system that turns meeting audio into auditable summaries, action items, owners, deadlines, and review decisions.

NLP FINAL PROJECT | PRODUCTION DESIGN | AGENTIC AI


**Course** Applications of Natural Language Processing in Industry

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PROBLEM DEFINITION

# Business problem: meetings create untracked commitments

Meeting decisions often stay inside audio, personal notes, or chat messages.
The business risk is not transcription; it is losing accountability after the meeting.

**1 Meeting audio**

Discussions, decisions and requests are mixed together.

**2 Manual notes**

Different people record different details; evidence is weak.

**3 Lost follow-up**

Owner, deadline or status becomes unclear.

**4 Business impact**

Time wasted, repeated clarification, missed commitments.

## Target users

Project managers, technical leads, product owners and team members who need traceable post-meeting tasks.

## Why NLP is required

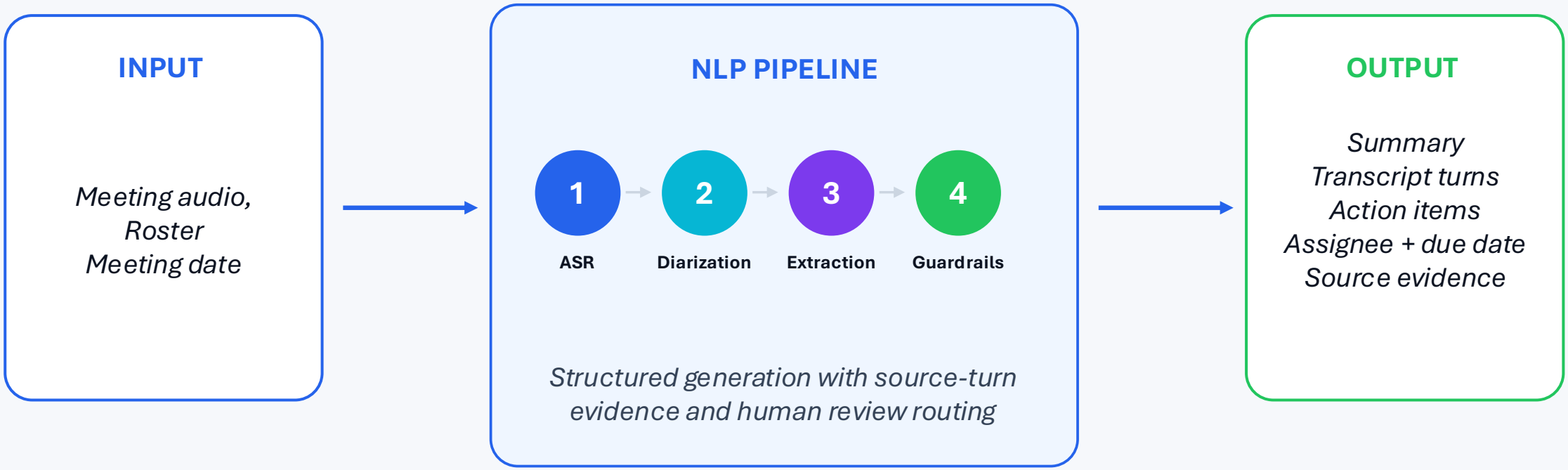
The system must infer commitments, assignees, relative deadlines and source evidence from natural conversation.

## Success criteria

Reduce manual minutes; improve task traceability; maintain precision, F1, schema stability and operational latency.

# Product objective: turn speech into accountable work items

The system is designed as a deployable product: *audio ingestion, asynchronous processing, structured output, review UI, persistence and monitoring.*




### Core claim

The value is the structured, auditable task object — not only a transcript.

### Production claim

Async jobs, database state and metrics make long-running NLP processing operationally feasible.

### Risk claim

Ambiguous or low-confidence outputs are routed to review instead of being silently trusted.

TASK DEFINITION

# NLP task formulation: from dialogue to verified JSON

The system decomposes one messy meeting into smaller NLP and data-engineering tasks, then validates the result before persistence.

### Example transcript turn

***“Bob, can you send the report by Friday?”***

**Required interpretation**

- detect commitment
- map alias “Bob” to roster
- normalize relative due date
- retain source turn for audit

### Structured action item

```
{
  "description": "send the report",
  "assignee": "Robert Kim",
  "due_date": "<meeting Friday>",
  "priority": "normal",
  "source_turn_ids": [..]
}
```

1 **ASR** *speech → text*

2 **Diarization** *speaker labels*

3 **Extraction** *summary & tasks*

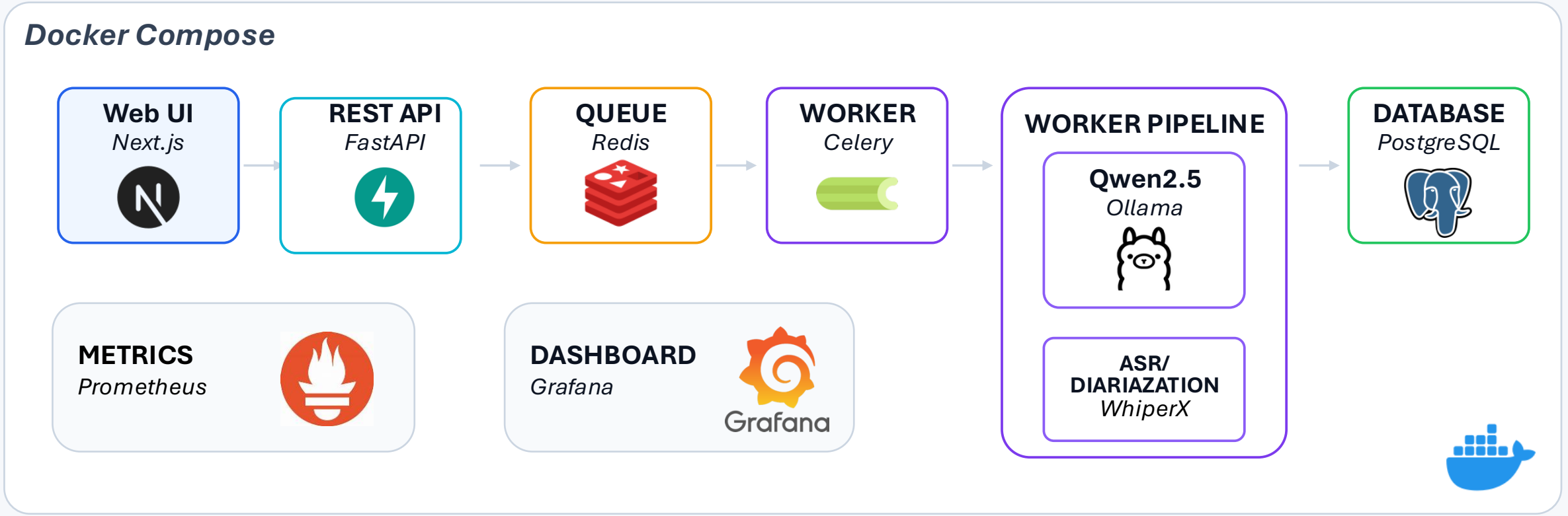
4 **Resolution** *assignee/date*

5 **Validation** *schema/evidence*

SYSTEM ARCHITECTURE

# Overview architecture

***Docker Compose*** separates the interactive product surface from heavy asynchronous NLP stages and observability.




### Why async?

ASR, diarization and local LLM inference exceed normal HTTP request latency; jobs need queueing and persistent state.

### Why local LLM?

Ollama/Qwen2.5-3B enables reproducible demo and reduces dependency on external APIs.

### Why monitoring?

Production review needs stage latency, failures, task counts and guardrail signals.

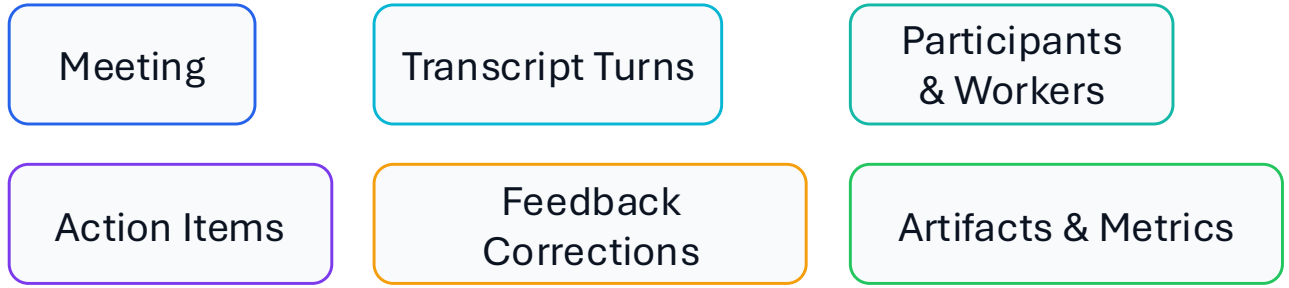
DATA

# Data management: synthetic benchmark + auditable schema

The report uses a reproducible synthetic benchmark while treating runtime audio/transcripts as sensitive data that must not be committed to Git.

| 205 | 2,036 | 1,620 / 416 | 60.01% | EN |
|---|---|---|---|---|
| meeting samples | transcript turns | train / test turns | positive turn rate | benchmark language |

## Data model




### Split strategy

Meeting-level 80/20 split prevents transcript turns from the same meeting leaking across train/test.

### Limitations

Synthetic benchmark is reproducible but does not fully cover accents, noisy audio, multilingual meetings or real privacy issues.

### Feedback data

User corrections become future evaluation/training data after PII scrubbing, validation and deduplication.

MODEL SELECTION

# Model design: LLM extraction with a real lightweight baseline

The production path extracts structured tasks; the baseline is intentionally smaller and solves action-turn detection as a reproducible comparison point.

## Main runtime pipeline

| | |
|---|---|
| **WhisperX** | ASR with timestamps |
| **Diarization** | speaker labels |
| **Qwen2.5-3B** | summary + JSON tasks |
| **Guardrails** | schema, PII, evidence checks |

## Traditional baseline

| | |
|---|---|
| **Input** | transcript turn text |
| **Features** | TF-IDF unigram/bigram |
| **Classifier** | Logistic Regression |
| **Task** | action-turn detection only |

EVALUATION

# Evaluation results: high precision, weaker recall and assignee resolution

The strongest result is stable JSON and low hallucination flags;
the main improvement target is assignee mapping and missed action items.

**LLM precision**

**0.8604**

*Most predicted tasks match reference items.*

**LLM recall**

**0.6665**

*Some reference action items are missed.*

**LLM F1**

**0.6886**

*Reported by benchmark script, not recomputed from rounded P/R.*

**Assignee accuracy**

**0.5232**

*Hard due to aliases, self-assignment and indirect ownership.*

**Schema failures**

**0.0%**

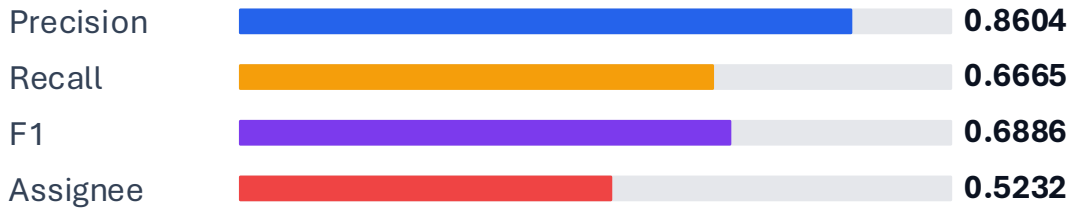
*JSON schema was stable on the benchmark.*

**Hallucination flags**

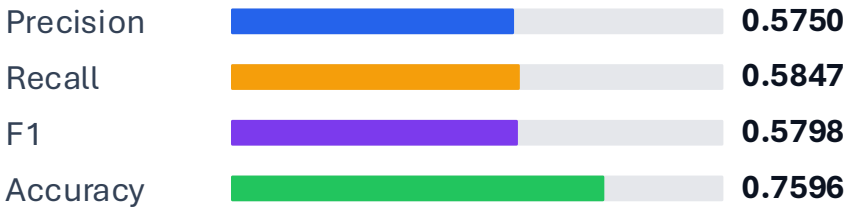
**0.0%**

*No task outside transcript by current heuristic.*

### LLM extraction metrics




### Baseline: action-turn detection




*Latency: average 26.97s; P95 120.2s → handled as asynchronous post-meeting processing.*

# Error analysis: where the system still needs engineering

The reported metrics point to precise but incomplete extraction.
The next iteration should focus on recall and ownership resolution.


**Indirect commitments**

"Maybe someone can…"
not an explicit task

**Assignee ambiguity**

alias, pronoun,
self-assignment

**Relative deadlines**

"next Friday"
needs meeting date

**Noisy transcript**

ASR error or
speaker overlap

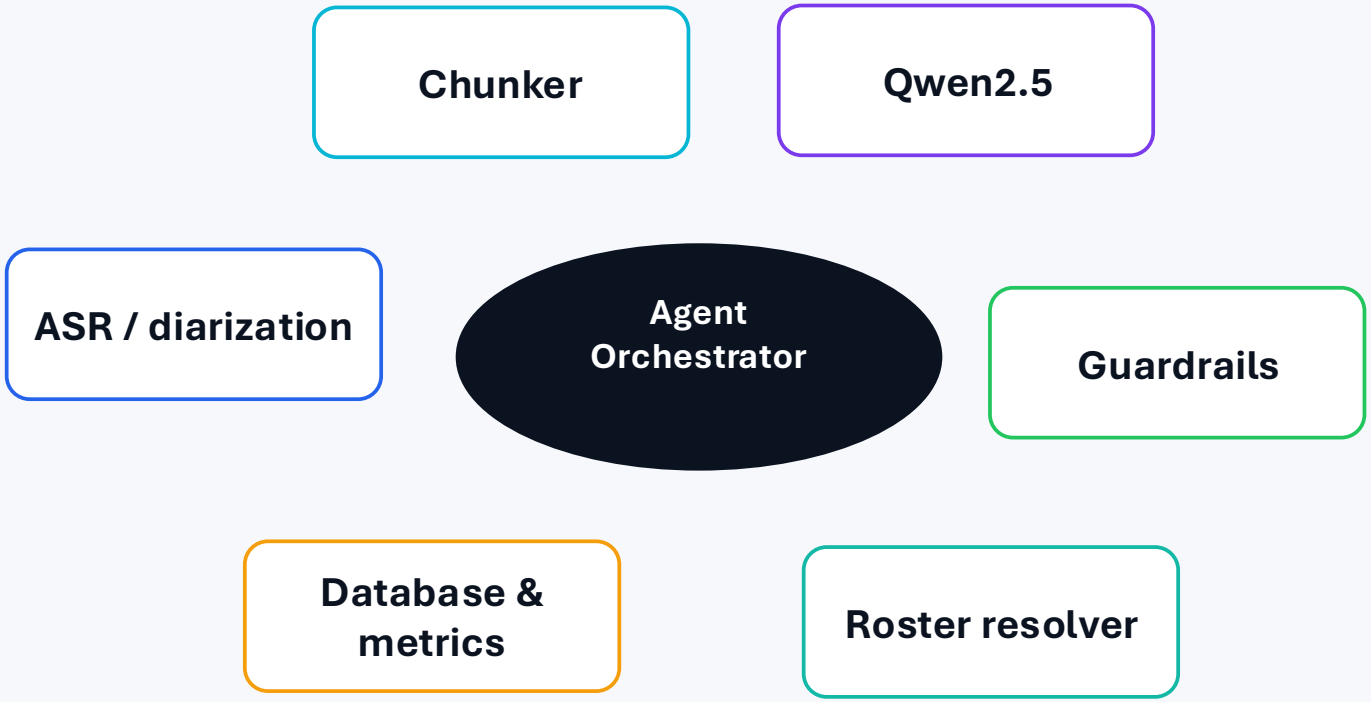
**Engineering response**

*keep source-turn evidence*
*improve roster/alias resolver*
*add stricter review routing*
*expand benchmark with real/noisy/multilingual meetings*

AGENTIC COMPONENT

# Agentic AI: decisions are made from intermediate signals

The agent is not a chatbot wrapper.
It orchestrates tools, validates intermediate outputs and decides whether results can be stored or must be reviewed.




**Decision rule example**

“Bob” uniquely matches roster alias → save Robert Kim. Ambiguous alias or low confidence → send to human review.

# Deployment design: asynchronous API, versioned models, review UI

The deployable interface is a REST/API + web demo over a background pipeline.
Users do not wait for LLM inference inside a single blocking request.

1. **POST /meetings**
   *upload audio + roster_json → returns meeting_id*
2. **Worker pipeline**
   *ASR, diarization, extraction, validation*
3. **GET /meetings/{id}**
   *status, transcript, summary, tasks, review items*
4. **POST /feedback**
   *user corrections saved for evaluation/training*

## Repository readiness

src/
data/
models/
configs/
tests/
pyproject.toml
README

## Operational constraints

Async jobs for latency
Redis/Celery for queueing
PostgreSQL for durable job state.

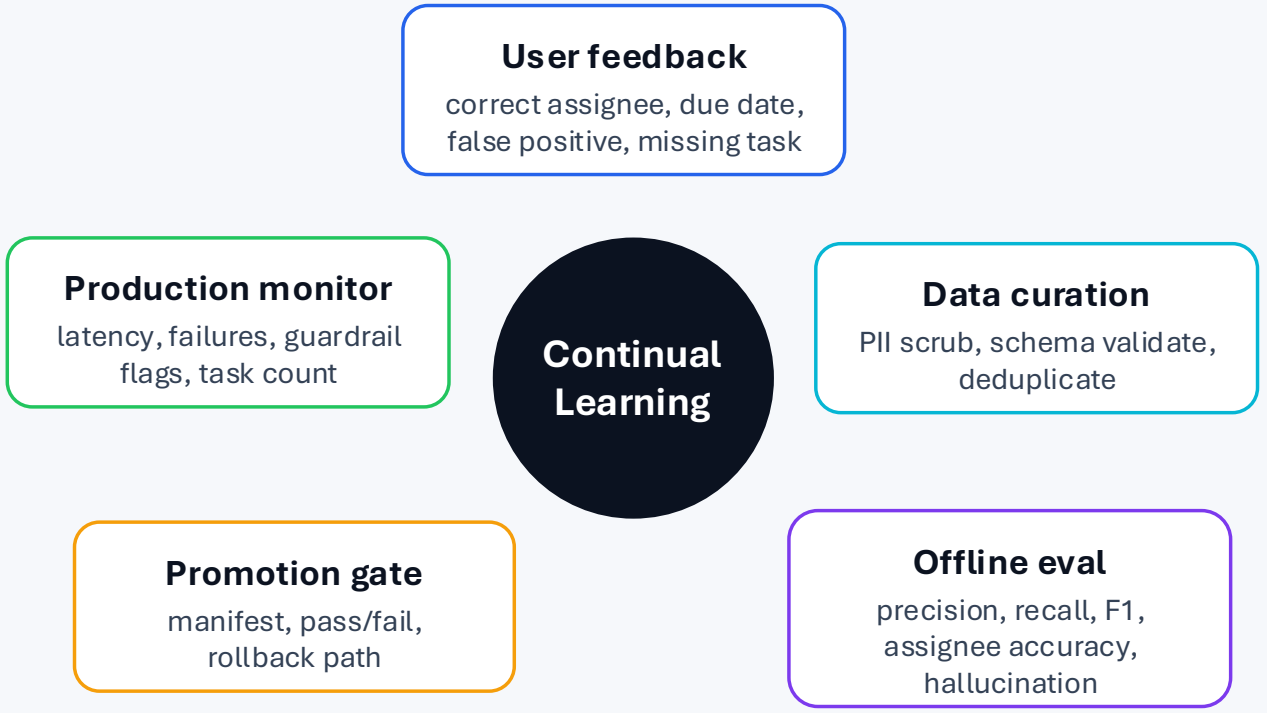
## Model governance

Model version stored
in meeting metadata.

MLOPS

# Monitoring and continual learning loop

New data should improve the system only after validation, PII scrubbing, benchmark evaluation and controlled promotion.




**Drift risks**

New domains, language mix, microphone quality and roster changes.

**Quality gates**

Candidate models must beat fixed benchmarks and preserve guardrail stability.

CONCLUSION

# Key takeaways and next iteration

The project satisfies the assignment as an end-to-end NLP system, but the honest engineering story is that production quality depends on review, monitoring and better real-world data.

### 1. End-to-end system

Audio → ASR/diarization → structured extraction → guardrails → database → web review → monitoring.

### 2. Evaluation is explicit

LLM extraction metrics and TF-IDF action-turn baseline are separated to avoid invalid comparisons.

### 3. Risk is handled by design

Ambiguity triggers review; outputs retain evidence; PII and misuse risks are documented.

## Future work roadmap

**Real data eval**
consent + anonymized meetings

**Audio robustness**
noise, overlap, accents

**Human-in-loop**
approval workflow + audit logs

***Final message: the system should be trusted as a decision-support assistant, not as an automatic source of truth.***

RESOURCES

# Resources & references

Project deliverables and academic references used throughout this work.

## PROJECT DELIVERABLES

**Data**
minhthien/mia-meeting

**Source Code**
nguyenvmthien/MIA

**Video Demo**
(add link)

## REFERENCES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